Changing Your Mind in Space

Personal AI for joint spatial work in XR

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The problem

I'm laying out a small exhibition in VR. I point at a panel and say, "Turn this towards the entrance, but keep that work visible." I sketch a visitor route, walk around the model, then drag a neighbouring exhibit somewhere else.

Moving that exhibit may quietly break the sight line I asked for, and the route I drew may still be a thought, not a decision. A collaborator who keeps up knows which parts of our plan just changed, which still hold, and what to do next.

My claim is that spatial work is a web of commitments, and changing one puts others in question. I propose to build an embodied AI that holds this web in the open, edits it with the person, and carries its consequences into action. I will measure how this changes the way person and AI repair misunderstandings and carry work across revisions and across days. The larger ambition is a personal AI with a shared history of work: one that remembers how decisions came to be and can help develop what comes next.

Research questions

RQ1. As people speak, point, draw and move objects together, how do they establish and revise what they are referring to, where things should go, and why?

RQ2. How does a visible, editable record linking objects, relations, reasons and next actions affect the reliability and effort of repairing misunderstandings with an embodied AI?

RQ3. When the goal or the scene changes between sessions, can the AI revise what the change touches and keep what still stands?

The system

The testbed is one Unity/OpenXR application with four to eight exhibits, a known entrance and a small set of spatial tools. People speak, point with a controller and sketch routes or relations. Gaze and free-hand input come later, if equipment allows. The agent has a consistent voice and a simple visible presence, held constant across conditions.

The Ledger. Each entry ties together an object, an intended spatial relation, the evidence for that reading, and the next proposed action. “Keep this visible from the entrance” becomes an exhibit, a vantage point and a sight line you can see and grab. Entries distinguish who supplied an interpretation, whether it is proposed or agreed, and whether it remains valid. A sketch enters as a proposal, and its role gets settled in conversation. Say “no, from the other door” or drag the vantage point, and the entry changes. The system follows the dependency links to flag affected actions and withdraw their previews, retaining unrelated work. Only the relevant entry sits beside the model, with the full overview available on demand.

Action. An off-the-shelf language model reads timestamped input and returns structured proposals. The assistant can highlight, draw, move, rotate, compare, explain and undo, through a finite tool API over known objects with stable identifiers. Geometry checks test overlap and line of sight. We log three things separately: what the model claimed, which tool calls succeeded, and whether the person’s intended relation actually holds.

Offline sequences test ambiguous references, moved objects, superseded instructions and scene changes mid-response, annotated independently. These tests guide the prototype before participant evaluation.

Study 1 How people build agreements in space

Twelve to sixteen participants with varied design experience work in pairs on layout and explanation tasks. They speak, point, draw and compare alternatives. Replay interviews ask how a remark becomes an agreement, how a sketch acquires its meaning, and how people notice and repair being misunderstood. A disclosed Wizard-of-Oz round then tries candidate AI responses and record designs.

Coding separates problems of reference, viewpoint, relation, intention and execution. Syiem et al.’s analysis of spatial dialogue [2] provides a starting point for examining how these actions connect. The outputs are a design space of coordination breakdowns, representative change events, and requirements for the prototype.

Study 2 Does the Ledger earn its place

A counterbalanced within-participant study will recruit approximately 24 to 30 participants; the pilot and a prespecified precision target will set the final number. Both conditions share the same model, task history, scene, inputs, agent presentation, highlights, previews, confirmation and undo. The baseline is a conversational assistant that can revise its reading through dialogue and inspectable summaries. The comparison tests the Ledger and its direct editing workflow.

Matched tasks include misunderstandings planted at comparable moments and disclosed as a manipulation: a wrong referent, a relation read from the wrong viewpoint, an action built on a superseded goal. A separate end-to-end task runs the live model without researcher correction. Planted and naturally occurring errors are analysed separately.

The primary outcome is successful repair before the first confirmed execution of the affected action, judged against an independently specified requirement. Repair time and clarification turns capture the effort involved. Further outcomes include unintended changes to agreed work, final constraint satisfaction, workload, perceived agency and social presence. Exchanges without errors are reported too. Mixed models account for repeated participants and tasks, with effect estimates and uncertainty.

Study 3 Picking the work back up

Twelve to sixteen participants make three visits over two to three weeks. Each visit involves arranging, inspecting and revisiting an exhibition model. Between visits something changes: a panel moves, the route is different, the visibility goal is revised.

Both conditions use the same Ledger interface and source history. A retrieval baseline reconstructs the relevant agreements and action links on return. The comparison mechanism maintains these links through change, flags those affected, and proposes local updates. Both use the same model and matched budgets. Task sequences are counterbalanced.

The primary measure is the proportion of assessable change opportunities in which the first executable proposal still depends on an invalidated object, relation or goal. Clarification requests and stalled tasks are reported alongside it. Resumption time, unnecessary revision of valid work, task quality and supervision effort show what each approach costs.

What the assistant remembers. Object references, agreed relations, rejected alternatives and their stated reasons, all timestamped. If an arrangement was rejected because it blocked a sight line, the AI keeps the reason and re-checks it against the new coordinates. Participants can inspect, correct and delete records. Temporary decisions remain distinct from standing preferences. Interviews explore continuity and whether people feel understood, read alongside memory errors and actual outcomes.

Where this sits

GazePointAR [3] and GesPrompt [4] connect speech with spatial cues. VRCopilot [5] co-creates layouts through editable intermediate representations. Eye2Eye [6] maintains evolving common ground; SpatialPrompt [9] turns sketch and voice into constraints for 3D generation. SpatiaLLM [10] verifies spatial memory against placement constraints, while MUSE [11] connects local scene editing with requirement tracking and verification.

These systems work out what I mean, generate what I ask for, or check what I placed. What remains open is what happens to the actions built on a decision when that decision changes, and who gets to see and fix those links. The Ledger puts those dependencies into the shared workspace, where they can be discussed and changed. I will use it to study a personal AI's ability to carry a joint project forward while its user is still working out what that project should become.

Why QUT and Türkay

The project grows from three threads in Selen Türkay's work. Ngo, Harman and Türkay [1] studied how people perceive the personalities of LLM-driven embodied agents in VR. Syiem, Türkay and colleagues [2] showed that drawing modality changes how people communicate about space. Gallagher, Türkay and colleagues [7] put immersive tools into real learning activity. My question sits where these meet: what does an embodied agent need to hold in order to take part in spatial communication as a task develops?

It also connects directly to the advertised QUT topic on intelligent agents in immersive systems [8]. Exhibition layout is my starting context. I would explore geological model interpretation with CLEVAR's collaborators as a further setting for shared spatial reasoning.

Research preparation

My training spans architecture and computer science. CAVE projection and interaction work taught me spatial layout and on-site system integration. Game Fellows began with rule-based companions for following, fetching and co-play, followed by a separate exploration of screenshot-based AI responses. This work shaped my interest in an assistant that contributes to a shared activity. My image-detection capstone taught me to audit outputs reproducibly and study individual failures. I will bring this combination of spatial practice, prototyping and evaluation to the XR system.

Plan

Year 1: Study 1 and an end-to-end prototype. Year 2: build and evaluate the Ledger and its repair workflow. Year 3: repeated use and change. Year 4: synthesis and follow-up. Studies 1 and 2 carry the thesis; Study 3 extends it through time. The project reuses existing models and a small asset library in a single application.

Ethics

Approval precedes recruitment. Recording covers task activity with informed consent, and participants can pause, rest or withdraw. External processing of speech or scene data is disclosed. Planted errors and Wizard-of-Oz behaviour are disclosed and logged. The system offers undo and control over retained task history.

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